Loan Approval Prediction using Machine Learning

Internship Task 2: Alfido Tech

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Abstract

This project aims to develop a machine learning model to predict whether a loan application will be approved based on applicant information. The dataset was preprocessed by handling missing values, encoding categorical variables, scaling numerical features, and balancing the training data using SMOTE. Two classification models, Logistic Regression and Decision Tree, were trained and evaluated using Precision, Recall, F1-Score, and ROC-AUC. The results showed that Logistic Regression achieved better performance than Decision Tree, making it the preferred model for loan approval prediction. This system can support financial institutions by improving the speed, consistency, and accuracy of loan approval decisions.

Objective

The main objective of this project is to build a machine learning model that accurately predicts loan approval status using applicant details. The project also aims to compare different classification algorithms and identify the best-performing model based on evaluation metrics.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Imbalanced-learn (SMOTE)
- Google Colab

Dataset

The Loan Prediction dataset contains information about loan applicants, including gender, marital status, education, employment status, applicant income, co-applicant income, loan amount, loan term, credit history, property area, and loan approval status. The target variable is Loan_Status, which indicates whether a loan is approved or not.

Data Cleaning

The following preprocessing steps were performed before model training:

- Removed and handled missing values using median and mode.
- Converted categorical variables into numerical values using Label Encoding.
- Scaled numerical features using StandardScaler.
- Split the dataset into training and testing sets.
- Applied SMOTE to balance the training dataset and reduce class imbalance.

Visualizations

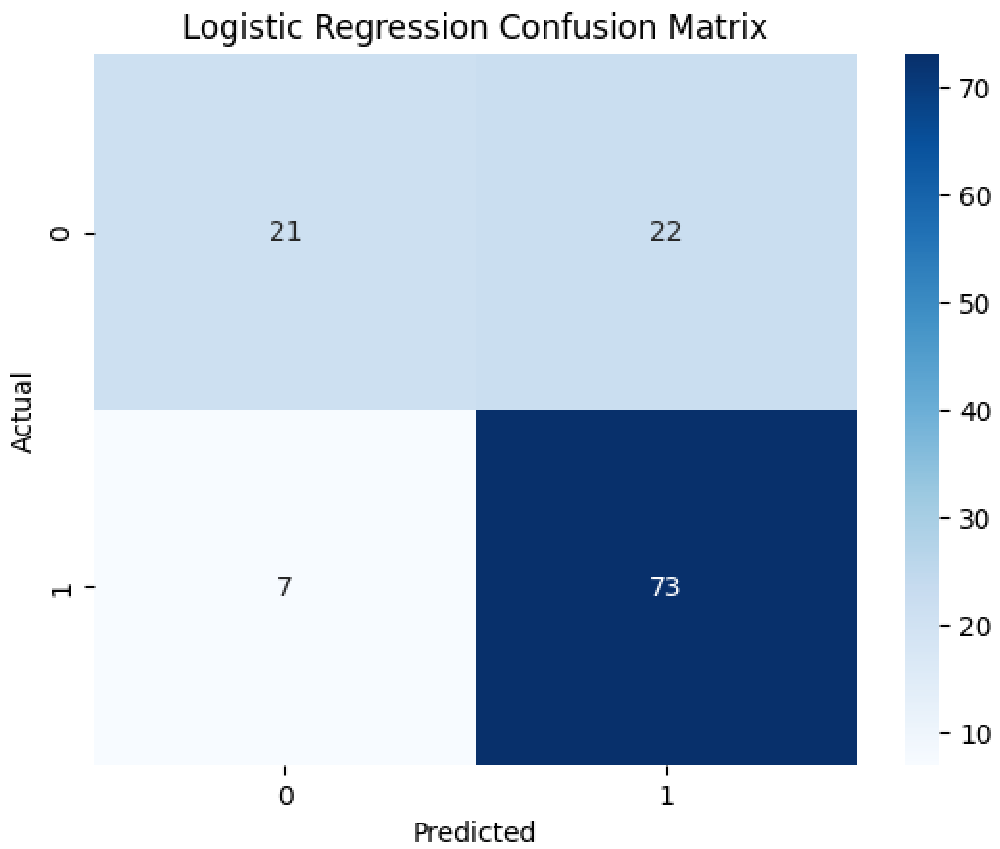

Fig 1.1: Logistic Regression Confusion Matrix

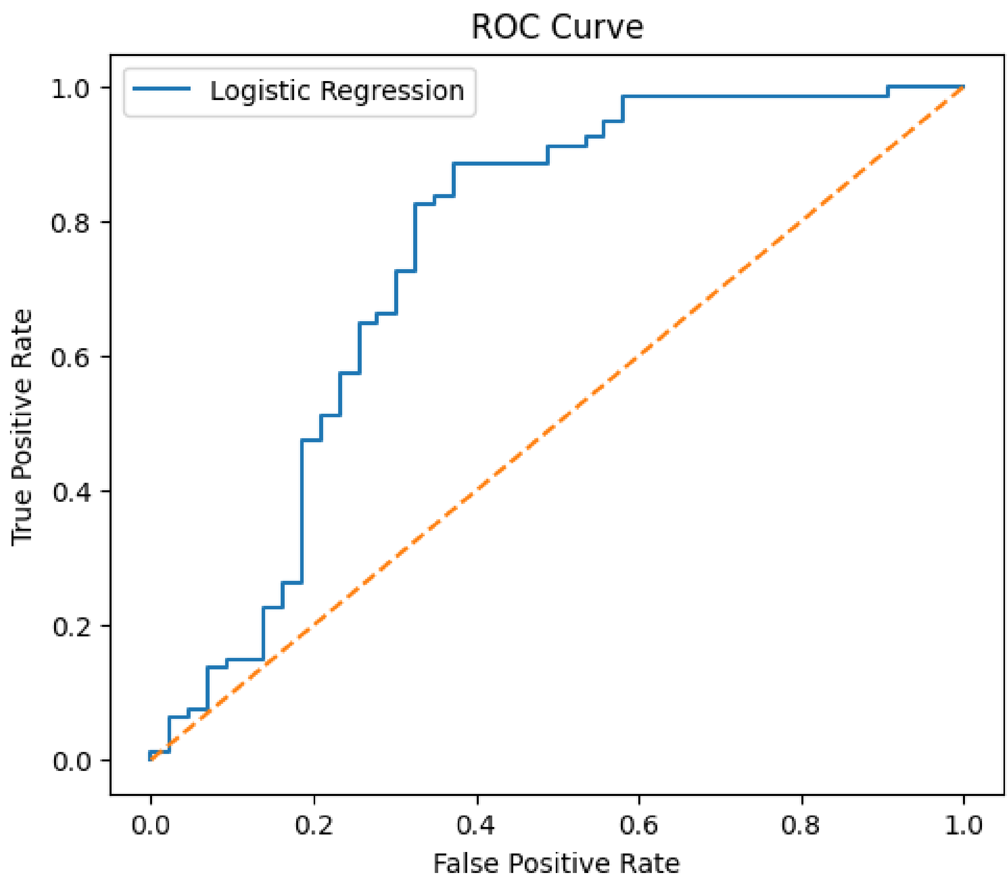

Fig 1.2:ROC Curve

Project Summary

This project focuses on predicting loan approval using machine learning techniques. The primary objective was to build a reliable classification model that helps financial institutions determine whether an applicant is eligible for a loan.

Initially, the dataset was imported into Google Colab and explored to understand its structure, missing values, and data types. Missing values were handled using appropriate statistical methods, while categorical variables were converted into numerical values using Label Encoding. Numerical features were standardized using StandardScaler to improve model performance.

Since the dataset contained class imbalance, the SMOTE (Synthetic Minority Over-sampling Technique) algorithm was applied to balance the training data. This helped improve the model's ability to classify both approved and rejected loan applications.

Two machine learning algorithms, Logistic Regression and Decision Tree Classifier, were trained and compared. Their performance was evaluated using Precision, Recall, F1-Score, and ROC-AUC. Confusion Matrix and ROC Curve visualizations were also generated to analyze classification performance.

The experimental results showed that Logistic Regression achieved higher Precision, Recall, F1-Score, and ROC-AUC compared to Decision Tree. Therefore, Logistic Regression was selected as the most suitable model for loan approval prediction.

This project demonstrates how machine learning can automate loan approval decisions, reduce manual work, improve decision-making consistency, and minimize financial risks for banking and financial institutions.

Key Insights

- Logistic Regression achieved better overall performance than Decision Tree.
- Credit history was one of the most influential factors affecting loan approval.
- Applying SMOTE improved the model's ability to classify minority class samples.
- Proper preprocessing significantly improved prediction performance.
- Machine learning can reduce manual loan verification time and improve decision accuracy.

Business Recommendations for Alfido Tech Platform

1.Automate Loan Screening: Integrate the prediction model into the platform to automatically screen loan applications before manual review.

2.Risk-Based Decision Making: Use prediction probabilities to classify applicants into low-risk and high-risk categories for better financial management.

3.Improve Customer Experience: Provide faster loan approval decisions, reducing waiting time and increasing customer satisfaction.

4.Continuous Model Monitoring: Regularly retrain the model with updated loan data to maintain prediction accuracy and adapt to changing customer behavior.

5.Data Quality Improvement: Encourage complete and accurate applicant information during data collection to further improve model performance.

Conclusion

This project successfully developed a machine learning model for loan approval prediction. After preprocessing the data and handling class imbalance using SMOTE, Logistic Regression and Decision Tree models were evaluated. Logistic Regression produced the best performance with higher Precision, Recall, F1-Score, and ROC-AUC. The developed model can support banks and financial institutions in making faster, more accurate, and consistent loan approval decisions while reducing operational costs and financial risks.